

Exercise – 5

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Deadline: 15.06.2026 – 1:59pm

Submission Instructions:

- Access E05 on cocalc.
- The exercise is given to you in the form of a jupyter notebook.
- Programming language: Python 3.10+
 - **You only need to modify the „solution.ipynb“ file.** More detailed instructions on where you need to insert your code can be found in this file.
 - Process all available data samples and build up a data structure according to the template given in the file. UUIDs should be unique for each entry. The timestamp field should be filled with an integer corresponding to a UNIX timestamp (UTC) converted from "yyyy-mm-dd hh:mm:ss.ms CEST" format.
 - For each timestamp, if data from multiple sources is available for the same field in the data structure, please compute the average.
 - The final cells in the solution.ipynb file grades your solution.
 - This will give you feedback on your solution, if any of the „assert“s receive an error, that indicates that you need to fix the indicated function(s).
- To submit your solution, simply save your, 'solution.ipynb' file.
- Grading will be conducted automatically by cocalc based on the evaluation in the notebook.

Task Description – IoT Sensing – Energy sellers

- Imagine a smart electric vehicle (EV) charging station that is connected to a solar panel. The goal of the smart charging station is to serve as many EVs with as much solar energy as possible. To do so, the smart charging station is interested in weather information (e.g., temperature, precipitation, cloudiness) from different sources. Bad (rainy, cloudy) indicates a lower output from the solar panel and thus less energy to charge EVs. However, lots of sun and a blue sky increase the solar panel output which allows the smart charging station to sell more energy.
- Your task is to create a program that queries (senses) the different data sources for weather information.
- The different sources provide different formats of weather data. Therefore, you need to transform it into a single data format for later processing. The `solution()` function has code snippets that can help you determine what the data from the weather stations look like.
- The expected output of your program is a combined set of weather data (that adheres to the defined format and structure) that contains all of the relevant weather data information from all available sensors.
- Voltage readings are from a TMP36 sensor and needs to be converted to temperature in °C before averaging.

Questions? : etce-etce@tu-clausthal.de